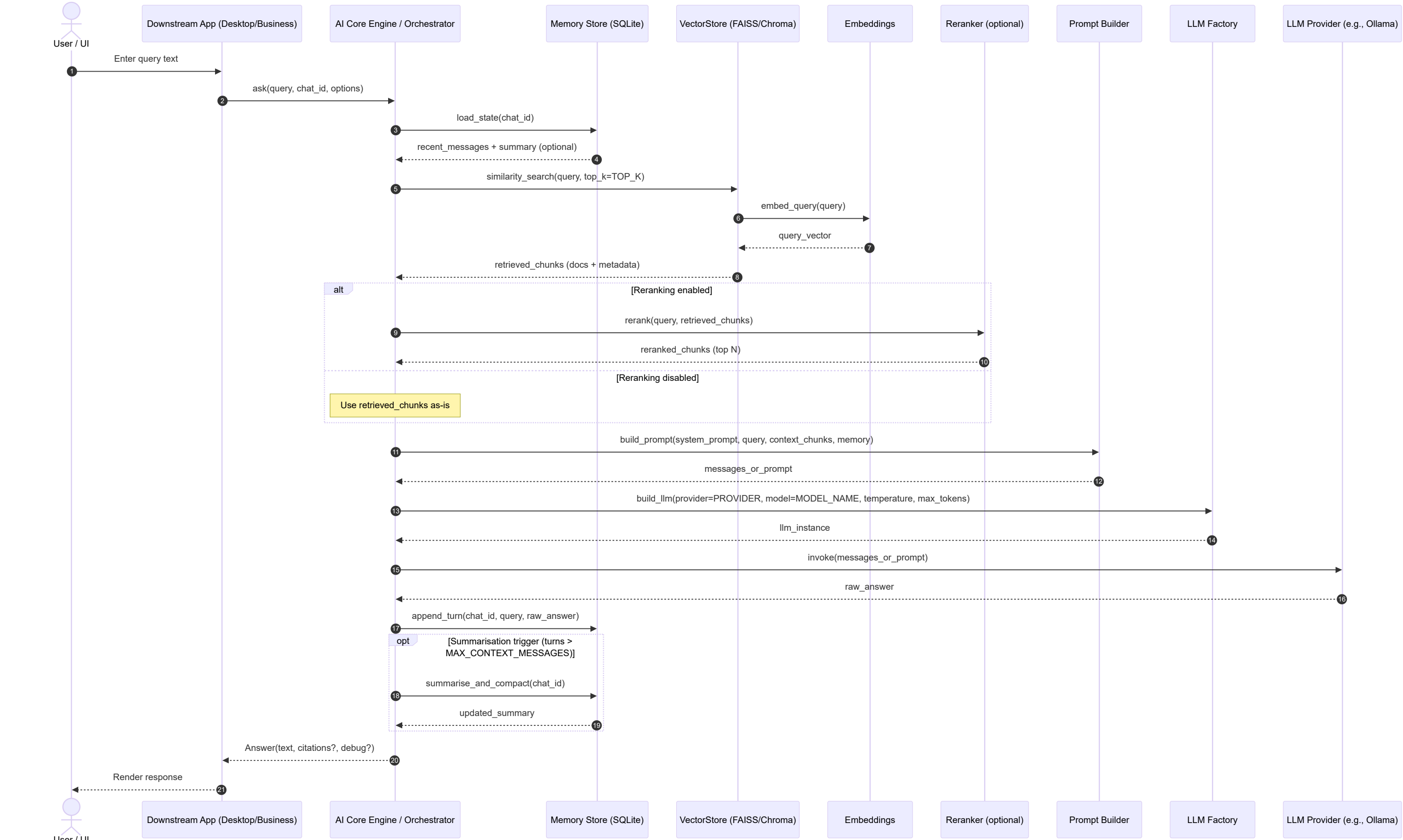


LifePIM AI Core sequence diagram

This page shows the full lifecycle of a single ask() call: from user query → retrieval → prompt assembly → LLM → response. (function names may change as code is tidied up)

Sequence diagram — ask() request lifecycle



The mermaid text for the above diagram is below

```
sequenceDiagram
autonumber
actor User as User / UI
participant App as Downstream App (Desktop/Business)
participant Eng as AI Core Engine / Orchestrator
participant Mem as Memory Store (SQLite)
participant VS as VectorStore (FAISS/Chroma)
```

participant Emb as Embeddings
participant RR as Reranker (optional)
participant PB as Prompt Builder
participant LLMF as LLM Factory
participant LLM as LLM Provider (e.g., Ollama)

User->>App: Enter query text
App->>Eng: ask(query, chat_id, options)

Eng->>Mem: load_state(chat_id)
Mem-->>Eng: recent_messages + summary (optional)

Eng->>VS: similarity_search(query, top_k=TOP_K)
VS->>Emb: embed_query(query)
Emb-->>VS: query_vector
VS-->>Eng: retrieved_chunks (docs + metadata)

alt Reranking enabled
Eng->>RR: rerank(query, retrieved_chunks)
RR-->>Eng: reranked_chunks (top N)
else Reranking disabled
Note over Eng: Use retrieved_chunks as-is
end

Eng->>PB: build_prompt(system_prompt, query, context_chunks, memory)
PB-->>Eng: messages_or_prompt

Eng->>LLMF: build_llm(provider=PROVIDER, model=MODEL_NAME, temperature, max_tokens)
LLMF-->>Eng: llm_instance

Eng->>LLM: invoke(messages_or_prompt)
LLM-->>Eng: raw_answer

Eng->>Mem: append_turn(chat_id, query, raw_answer)
opt Summarisation trigger (turns > MAX_CONTEXT_MESSAGES)
Eng->>Mem: summarise_and_compact(chat_id)
Mem-->>Eng: updated_summary
end

Eng-->>App: Answer(text, citations?, debug?)
App-->>User: Render response